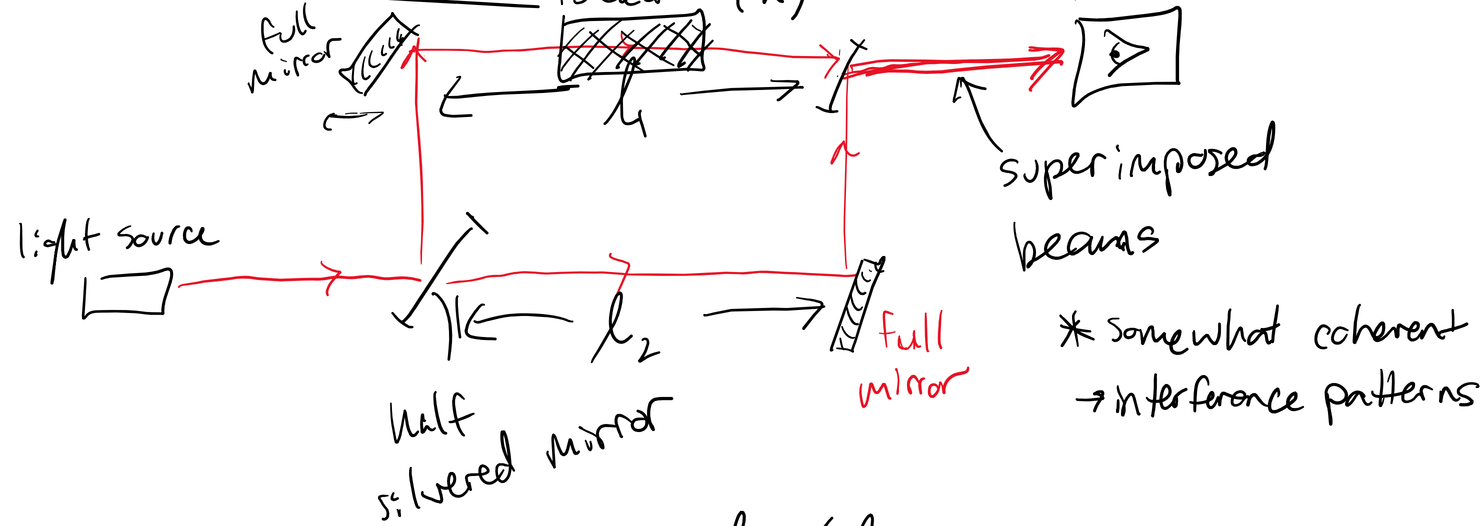
Basic Interferometer



$l_1 \neq l_2$ → interference changes

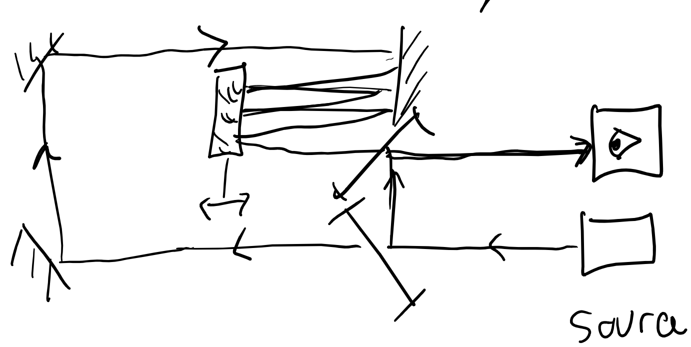
$$\Phi \Rightarrow \text{oscillation phase} = \frac{2\pi}{\lambda_0}(l_1 - d) + \frac{2\pi n}{\lambda_0} - \frac{2\pi l_2}{\lambda_0}$$

$$\Delta\Phi = \frac{2\pi}{\lambda_0}(\Delta n d)$$

↑ change in index of refraction

$$\Delta\Phi = \frac{2\pi}{\lambda_0}(\Delta l) \; ; \; \Delta l = |l_1 - l_2|$$

Modified to detect changes in length



ether → aether → æ
"ee"

Vacuum

$\varepsilon_0 = 8.85\,pF/m$

$\mu_0 = 4\pi \times 10^{-7}\ H/m$

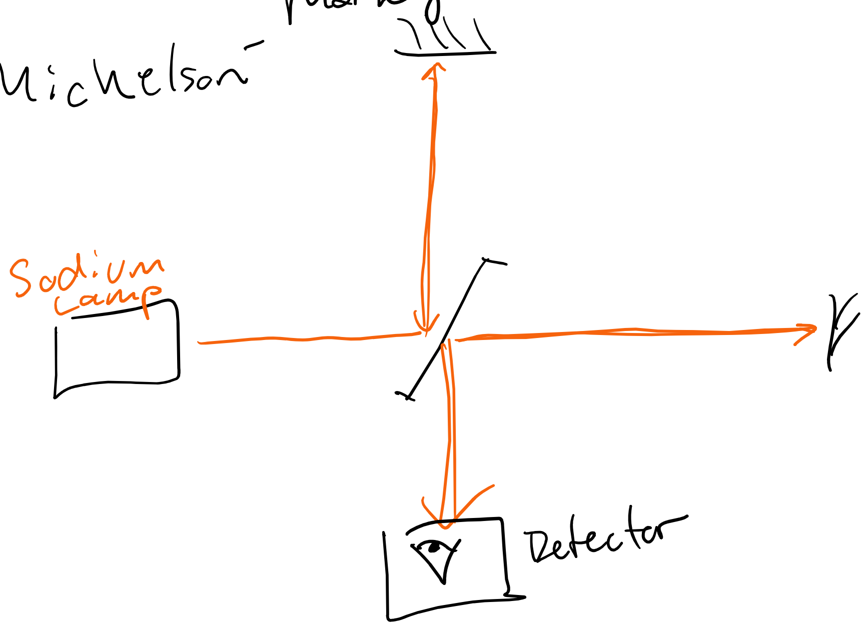
$\approx 1.2\,\mu H/m$

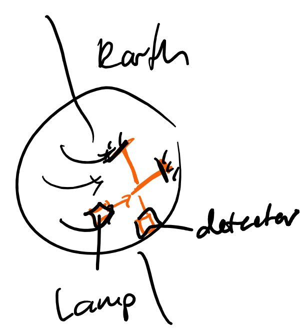
2 theory

1.) Static aether →

2.) Dynamic aether

Michelson-Morley





*null (didn't measure anything)

Sagnac